

5. BENDING STRESSES IN BEAMS

5.1 Introduction

Due to external loads and moments bending moments and shearing forces are set up at all the sections of a beam. A beam section has to resist the action of bending moment and shearing force so induced and the beam deforms. The process of bending will stop when it has set up full resistance to the bending moment and shearing force.

The stress produced at the section to resist the bending moment is known as bending stress or direct stress or flexural stress and that to resist the action of shear force is known as shearing stress.

5.2 Theory of simple bending

Assumptions: Before developing elastic flexural formula it will be worthwhile to outline the assumptions upon which such relationship depends:

- (1) The material of the beam remains elastic during loading i.e. elastic limit is not exceeded.
- (2) The material of the beam is perfectly homogenous and isotropic and modulus of elasticity in tension is equal to that in compression.
- (3) Before loading takes place the beam is straight and each cross section is symmetric about plane of bending.
- (4) Plane cross section remains plane before and after bending.
- (5) There is no resultant force perpendicular to any cross section.
- (6) Every layer of material is free to expand or contract longitudinally and laterally under stress and does not exert pressure upon each other.

Pure Bending or Simple Bending:

Fig. 5.1 shows a beam AB with supports at C&D, L units apart. Let the overhangs AC = BD = a. Let a point load be applied at each end of the beam. It can be observed that there is a constant bending moment and no shear force (zero shear force) between C&D. The stresses will be set up in that length of the

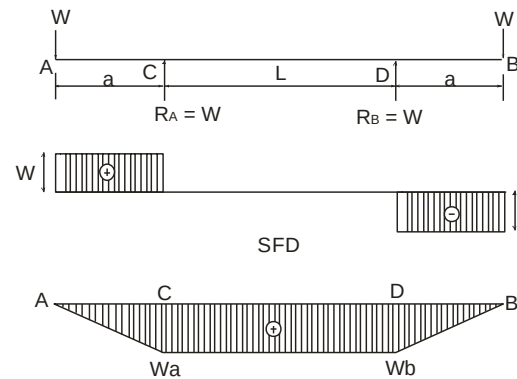


Fig. 5.1

beam due to bending moment only and that length of the beam is said to be in pure bending or simple bending.

Fig. 5.2 Shows the longitudinal section of a beam through plane of symmetry before and after it has been bent by two equal and apposite moments applied at its ends. The axes of coordinates are also shown in the same figure. The bent form of the longitudinal beam section shown in Fig. 5.2 (b) is highly exaggerated but it represents the essential features.

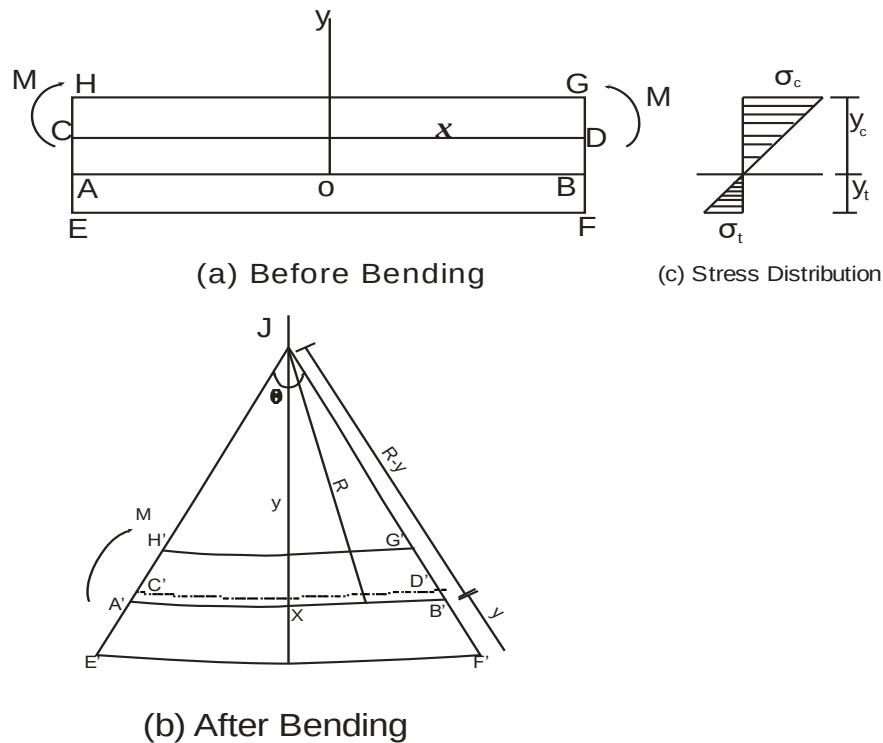


Fig. 5.2

Since the bending moment at any section is same, the section of the beam under consideration will bend uniformly or the curvature of any surface will be uniform as in Fig. 5.2 b. Thus the sides of longitudinal section GH and EF will bend into concentric arcs of circles, $G'H'$ and $E'F'$. All surfaces between and parallel to GH and EF will bend into contracted arcs of circles with centre at J . From Fig. 5.2(b) it can be seen that while $G'H'$ has contracted from its original length GH , the lower fiber $E'F'$ has extended from its original length EF . If we go down from $G'H'$ to $E'F'$, the contraction reduces becomes zero at some layer and then the layer begins to extend.

The layer of beam at which the extension or contraction does not take place, i.e. the layer which remains in its original length after bending is termed as or neutral layer. Line

AB in Fig. 5.2(a) and arc A'B' in fig. 5.2(b) represent neutral layer. The line of intersection of neutral layer and beam's cross section is defined as neutral axis.

The Fig. 5.2(b) shows the position of beam after bending. Let the radius of the neutral layer after and before bending be R so that the radius of layer C'D' is 'R-y'. It may be noted that AB = CD before bending and AB = A'B' after bending being neutral layer. C'D' has shortened from its original length CD. If the angle subtended by concentric arcs at common centre J is θ then

$$AB = CD = A'B' = R \theta$$

and $C'D' = (R - y) \theta$

strain of fiber CD $e = \frac{\text{Change in length}}{\text{Original Length}} = \frac{C'D' - A'B'}{A'B'}$

$$e = \frac{(R - y)\theta - R\theta}{R\theta} = \frac{-y}{R} \quad (\text{compressive}) \quad \dots\dots(5.1)$$

The negative sign indicates that strain is compressive. If another layer at a distance y below the neutral plane is considered, e will be positive or tensile.

Equation $e = \frac{-y}{R}$ shows strain varies directly as the distance from neutral axis

because R is constant for a given loading.

Since stress = strain x elasticity

$$\sigma = e \times E = \frac{y}{R} E$$

$$\frac{\sigma}{y} = \frac{E}{R} \quad \dots\dots (5.2)$$

or $\sigma \propto y$

$\sigma_t \propto y_t$

$\sigma_c \propto y_c$

Hence the stress intensity in any fiber is proportional to the distance of the fiber from the neutral layer. For the case explained above all fibers below the neutral layer are subjected to tensile stresses while those above the neutral layer are subjected to compressive stresses. The stress distribution diagram is shown in Fig. 5.2(c).

Moment of Resistance:

Let us now determine the magnitude of stress from the equation of equilibrium, since there is no external force, the sum of all the compressive forces above the neutral layer must be equal to the sum of the tensile forces below the neutral layer.

Let us take a strip δy thick at a distance y from NA in Fig 5.3 and let its area be δa . Since the stress is equal to ' σ ' there the thrust exerted by this small area will be ' $\sigma \cdot \delta a$ '.

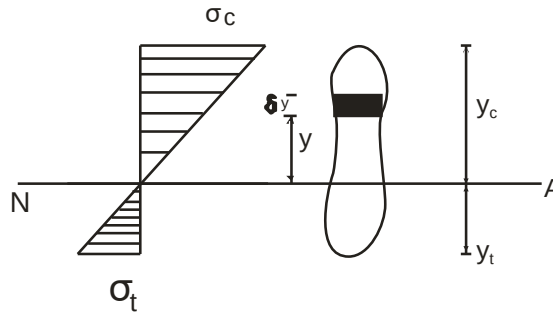


Fig. 5.3 Stress Distribution

To find M_r let us again consider the elementary area δa in Fig. 5.3. The force set up by the area $\sigma \delta a$ and the moment of force about NA is

$$\delta M = \sigma \cdot \delta a \cdot y = \frac{E}{R} (\delta a) \cdot y^2$$

The total moment of resistance set up by such elementary area will be

$$\sum_{y=y_t}^{y=y_c} \frac{E}{R} \delta a \cdot y^2 = \frac{E}{R} \int_{y_t}^{y_c} y^2 \delta a$$

The expression in the integral is known as second moment of area or moment of inertia about NA and is shown as 'I'

Hence $M_r = \frac{E}{R} I$

for equilibrium $M = M_r = \frac{E}{R} I$

$$\frac{M}{I} = \frac{E}{R} \quad \text{.....(5.4)}$$

But we know $\frac{E}{R} = \frac{\sigma}{y}$

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{y} \quad \text{.....(5.5)}$$

$$\frac{M}{I} = \frac{\sigma}{y}$$

For tensile side use σ_t and y_t

$$\frac{M}{I} = \frac{\sigma}{y}$$

For compressive side use σ_c and y_c

'EI' is termed as *flexural rigidity* and may be specified as a single constant for a given beam.

Eq. 5.5 is applicable to a member which is subjected to a constant bending moment and the member is absolutely free from shear force but in actual practice, a member is subjected to such loading that the B.M. varies from section to section and also the shear force is not zero. But shear force is zero at a section where bending moment is maximum. Hence the condition of simple bending may be assumed to be satisfied at such a section.

5.3 Design Criterion And Section Modulus

The bending equation is quite significant while designing machine components. To ensure strength in a beam it is necessary that the greatest tensile and compressive bending stress at the critical section i.e. where bending moment is maximum should not exceed the corresponding allowable stress for the material.

$$\sigma_{tmax} = \frac{M}{I} y_t ;$$

$$\sigma_{cmax} = \frac{M}{I} y_c$$

Condition of strength or of design

$$\sigma_{tmax} \leq \sigma_t \text{ allowable and } \sigma_{cmax} \leq \sigma_c \text{ allowable}$$

for a given value of allowable stress

$$\sigma = \frac{M}{I} y$$

$$M = \sigma \frac{I}{y} = \sigma Z \quad [\text{Compare with } F = \sigma.A \text{ for tensile loading}] \dots\dots(5.6)$$

Where $Z = \frac{I}{y}$ and is known as *section modulus*. Greater the value of Z stronger will be the section.

5.4 Section Modulus For Various Shapes Of Beam Section :

5.4.1 Rectangular Section

Moment of inertia of a rectangular section (Fig 5.4) about an axis through its C.G. (or through NA) is given by,

$$I = \frac{bd^3}{12}$$

Distance of outer most layers from N.A. is given by,

$$y_{\max} = \frac{d}{2}$$

∴ Section modulus is given by

$$Z = \frac{I}{y_{\max}} = \frac{\frac{bd^3}{12}}{\left(\frac{d}{2}\right)}$$

$$= \frac{bd^3}{12} \times \frac{2}{d} = \frac{bd^2}{6}$$

.....(5.7)

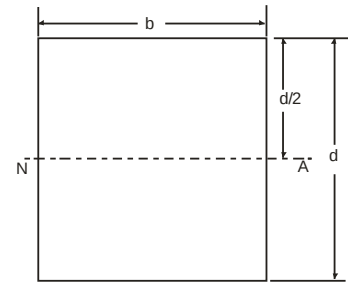


Fig. 5.4

5.4.2 Hollow Rectangular Section

Moment of inertia of a hollow rectangular section (Fig 5.5) about an axis through its C.G. (or through NA) is given by,

Here $I = \frac{BD^3}{12} - \frac{bd^3}{12}$

$$= \frac{1}{12} [BD^3 - bd^3]$$

$$\begin{aligned} y_{\max} &= \frac{D}{2} \\ \therefore Z &= \frac{I}{y_{\max}} = \frac{\frac{1}{12} [BD^3 - bd^3]}{\left(\frac{D}{2}\right)} \\ &= \frac{1}{6D} [BD^3 - bd^3] \end{aligned}$$

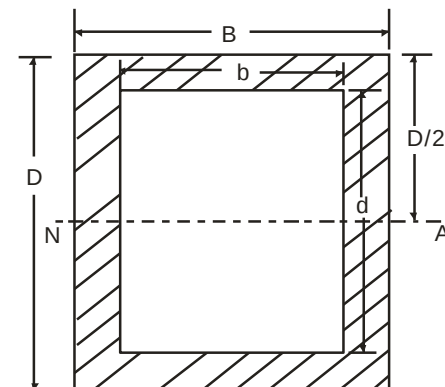


Fig. 5.5

$$= \frac{1}{6D} [BD^3 - bd^3]$$

.....(5.8)

5.4.3 Circular Section

Moment of inertia of a circular section (Fig 5.6) about an axis through its C.G. (or through NA) is given by,

$$I = \frac{\pi}{64} d^4$$

and

$$y_{\max} = \frac{d}{2}$$

$$\frac{I}{y_{\max}} = \frac{\frac{\pi}{64} d^4}{\left(\frac{d}{2}\right)}$$

∴

$$Z =$$

$$= \frac{\pi}{32} d^3$$

.....(5.9)

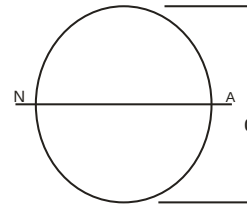


Fig. 5.6

5.4.4 Hollow Circular Section

Moment of inertia of a hollow circular section (Fig 5.7) about an axis through its C.G. (or through NA) is by,

Here $I = \frac{\pi}{64} [D^4 - d^4]$

and

$$y_{\max} = \frac{D}{2}$$

$$\begin{aligned} \therefore Z &= \frac{I}{y_{\max}} = \frac{\frac{\pi}{64} [D^4 - d^4]}{\left(\frac{D}{2}\right)} \\ &= \end{aligned}$$

$$= \frac{\pi}{32D} [D^4 - d^4]$$

.....(5.10)

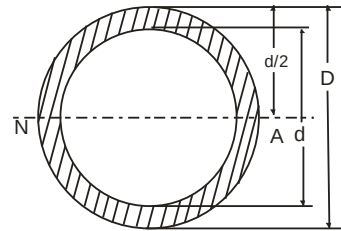


Fig. 5.7

5.7)
given